

Each time you step foot inside a casino, the probability of coming out a winner is pretty low. You shouldn't be surprised; after all, we all know the house has the best odds. But one game, if played correctly, gives you a legitimate chance to beat the house: blackjack. In a worldwide best seller called *Beat the Dealer: A Winning Strategy for the Game of Twenty-One*, Edward O. Thorp, a mathematician by training, outlined a process for outsmarting the casino.¹⁷

Thorp's strategy was based on a simple concept. When the deck is rich with 10s, face cards, and aces, the player—let's say it's you—has a statistical advantage over the dealer. If you assign a -1 for the high cards and a $+1$ for the low cards, it's quite easy to keep track of the cards dealt; just keep a running tally in your head, adding and subtracting as each card shows. When the count turns positive, you know there are more high cards yet to be played. Smart players save their biggest bets for when the card count reaches a high relative number.

Buried deep inside Thorp's book was a notation on the Kelly betting model.¹⁸ Kelly, in turn, took his inspiration from Claude Shannon, the inventor of information theory.

A mathematician with Bell Laboratories in the 1940s, Shannon spent a good deal of his career trying to find the most optimal way to transmit information over copper lines without having the information become garbled by random molecular noise. In 1948, in an article called "A Mathematical Theory of Communication," he described what he had discovered.¹⁹ Included in the article was the mathematical formula for the optimal amount of information that, considering the possibility of success, can be pushed through a copper wire.

A few years later, J. L. Kelly, another mathematician, read Shannon's article and realized the formula could just as easily work in gambling—another human endeavor that would be enhanced by knowing the possibilities of success. In 1956, in a paper titled "A New Interpretation of Information Rate," Kelly pointed out that Shannon's various transmission rates and possible outcomes of a chance event are essentially the same thing—probabilities—and the same formula could optimize both.²⁰

The Kelly optimization model, often called the optimal growth strategy, is based on the concept that if you know the probability of success, you bet